Date: 17-06-2025

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Mr. Vivan Mendonca** (Employee No: **NITK/CSD/153**) was employed with the **Centre for System Design Lab, National Institute of Technology Karnataka (NITK),** under the **E-Mobility Sector.** He began his tenure as a **Research Volunteer** on **July 1, 2024,** and transitioned to the role of **Junior Research Fellow (JRF) on September 19, 2024, a role he held until June 16, 2025.**

Mr. Vivan consistently demonstrated exceptional dedication, contributing **45 hours per week** to innovative projects in electric mobility and autonomous systems.

Mr. Vivan demonstrated strong technical expertise and problem-solving abilities in autonomous systems and electric vehicle development. His contributions have significantly advanced our E-Mobility initiatives through practical implementation and innovation.

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His key projects and technical contributions include:

1.Assistive Mobility Systems Design and Prototyping:

- Designed and prototyped electric wheelchairs for persons with disabilities (PwD) with advanced mobility features using Boeing grants
- Integrated safety mechanisms and control systems, ensuring reliability and ease of operation in varied conditions
- Developed a self-locking mechanism and a universal wheelchair frame to enhance adaptability across user requirements
- Optimized manoeuvrability and stability through kinematic and dynamic analysis
- Used 3D printing and laser cutting for rapid prototyping of components, enabling iterative design improvements
- Integrated a GPS tracking system for navigation and location monitoring using TB Track platform

2. Autonomous Vehicle Mechanical and Electrical Systems Integration:

- Transforming the Mahindra REVA electric vehicle into an autonomous campus shuttle
- Designed and fabricated the REVA electric vehicle to support autonomous vehicle development
- Performed structural analysis using Ansys for optimized mechanical-electrical integration
- Installed a new electric drive differential with its controller, battery, and integrated electronics
- Designed and analysed vehicle components using SolidWorks, CAD and FEA
- Modified chassis and swing arms to support the electric systems

3. Autonomous Vehicle Systems Development:

- Leading Mahindra REVA autonomous shuttle development integrating LiDAR, cameras, and IMUs
- Implementing Autoware-based autonomous navigation stack with ROS2 framework
- Developing collision avoidance algorithms and safety systems using Python
- Working on perception, localization, and navigation systems integration
- Designing and implementing real-time control systems using MATLAB/Simulink

4. Electric Vehicle Overhaul, Modification, and Maintenance:

- Led the overhaul of the electric trike, including rewiring the electrical system and replacing components such as motors and controllers
- Implemented design changes to improve manoeuvrability and ensured system functionality using industry-standard tools
- Performed maintenance and repair of various electric vehicles, including e-bikes, e-cycles, trikes, and scooters
- Diagnosed issues, replaced worn-out components, and ensured optimal vehicle performance and safety
- Documented maintenance activities for long-term reference and quality assurance

5. RF Antenna Development and Emergency Communication:

- Designed, fabricated, and optimized 6-meter Yagi-Uda antennas for high-gain performance
- Applied advanced electromagnetic principles for antenna theory and disaster management applications
- Built a 2-meter Yagi beam antenna used in fox hunting for designing lifeline antennas to enhance emergency communication
- Participated in mastering high-distance communication unlocking techniques in amateur radio through dedicated project work
- Participated in workshops, including JOTA (Jamboree on the Air), for disaster response readiness
- Participated in SEARCH (System for Emergency Assistance, Response, and Communication Hub) activities related to emergency response solutions

Technical Environment:

CAD/CAE: SolidWorks, Ansys

Software: Autoware/ROS2, Python, MATLAB/Simulink, Ubuntu/Linux

Hardware: LiDAR, IMU sensors, Electric drive systems, Embedded systems

Vivan's work during this period has been focused on sustainable mobility and real-time system design under the E-Mobility Sector.